

Администрирование сетевых подсистем

Настройка NFS (Лабораторная работа № 13)

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки настройки сервера NFSv4 для удалённого доступа к ресурсам и организации общего сетевого хранилища.

- Настроить экспорт каталогов на сервере NFS
- Обеспечить корректную работу SELinux-контекстов
- Настроить firewalld для работы NFS и RPC-служб
- Подключить ресурсы на клиенте и настроить автомонтирование
- Включить в дерево NFS каталог веб-сервера и пользовательские каталоги
- Автоматизировать настройку через provision-скрипты Vagrant

Выполнение лабораторной работы

Экспорт корня дерева NFS

```
[vagrant@server ~]$ dnf -y install nfs-utils
Error: This command has to be run with superuser privileges (under the root user on most systems).
[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install nfs-utils
Last metadata expiration check: 1:21:33 ago on Thu 12 Mar 2026 09:05:56 AM UTC.
Dependencies resolved.
```

Package	Arch	Version	Repository	Size
Installing:				
nfs-utils	x86_64	1:2.5.4-38.el9_7.3	baseos	434 k
Upgrading:				
libipa_hbac	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libldb	x86_64	4.22.4-12.el9_7	baseos	180 k
libsmbclient	x86_64	4.22.4-12.el9_7	baseos	73 k
libsss_certmap	x86_64	2.9.7-4.el9_7.1	baseos	88 k
libsss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	39 k
libsss_nss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	43 k
libsss_sudo	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libtalloc	x86_64	2.4.3-1.el9	baseos	33 k
libtdb	x86_64	1.4.13-1.el9	baseos	53 k
libtevent	x86_64	0.16.2-1.el9	baseos	50 k
libwbclient	x86_64	4.22.4-12.el9_7	baseos	42 k
samba-client-libs	x86_64	4.22.4-12.el9_7	baseos	5.3 M
samba-common	noarch	4.22.4-12.el9_7	baseos	173 k
samba-common-libs	x86_64	4.22.4-12.el9_7	baseos	104 k
sssd	x86_64	2.9.7-4.el9_7.1	baseos	25 k
sssd-ad	x86_64	2.9.7-4.el9_7.1	baseos	217 k
sssd-client	x86_64	2.9.7-4.el9_7.1	baseos	158 k
sssd-common	x86_64	2.9.7-4.el9_7.1	baseos	1.6 M
sssd-common-pac	x86_64	2.9.7-4.el9_7.1	baseos	94 k
sssd-ipa	x86_64	2.9.7-4.el9_7.1	baseos	286 k
sssd-kcm	x86_64	2.9.7-4.el9_7.1	baseos	107 k
sssd-krb5	x86_64	2.9.7-4.el9_7.1	baseos	70 k
sssd-krb5-common	x86_64	2.9.7-4.el9_7.1	baseos	92 k
sssd-ldap	x86_64	2.9.7-4.el9_7.1	baseos	159 k
sssd-proxy	x86_64	2.9.7-4.el9_7.1	baseos	70 k

SELinux и запуск NFS-службы

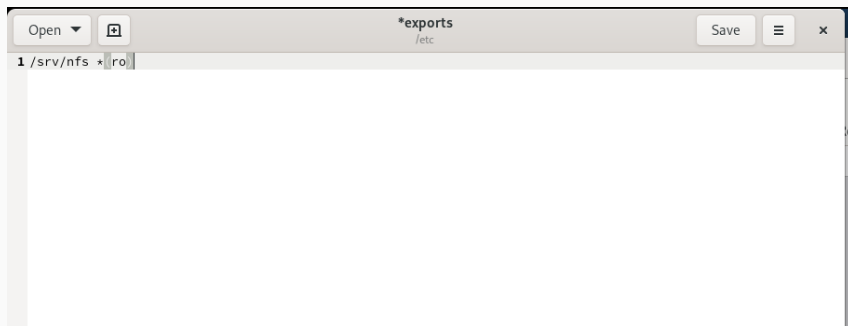


Рис. 2: Настройка SELinux-контекста и запуск службы NFS

Настройка firewalld для RPC-служб

```
(gedit:7260): dconf-WARNING **: 10:35:53.460: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server etc]# ^C
[root@server etc]#
[root@server etc]#
[root@server etc]# semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
[root@server etc]# restorecon -vR /srv/nfs
Relabeled /srv/nfs from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:nfs_t:s0
[root@server etc]# systemctl start nfs-server.service
[root@server etc]# systemctl enable nfs-server.service
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/
system/nfs-server.service.
[root@server etc]# firewall-cmd --add-service=nfs
bash: firewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=nfs
success
[root@server etc]# firewall-cmd --add-service=nfs --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]#
```

Рис. 3: Добавление служб mountd и rpc-bind в firewalld

Проверка экспортов с клиента

```
root@server:/etc

[root@server etc]# sof | grep TCP
bash: sof: command not found...
[root@server etc]# lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
       Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
       Output information may be incomplete.
systemd      1                root    62u    IPv4        46424      0t0      TC
P *:sunrpc (LISTEN)
systemd      1                root    64u    IPv6        45670      0t0      TC
P *:sunrpc (LISTEN)
cupsd        953                root     6u    IPv6        21155      0t0      TC
P localhost:ipp (LISTEN)
cupsd        953                root     7u    IPv4        21156      0t0      TC
P localhost:ipp (LISTEN)
sshd         968                root     3u    IPv4        21197      0t0      TC
P *:down (LISTEN)
sshd         968                root     4u    IPv6        21199      0t0      TC
P *:down (LISTEN)
sshd         968                root     5u    IPv4        21201      0t0      TC
P *:ssh (LISTEN)
sshd         968                root     6u    IPv6        21203      0t0      TC
P *:ssh (LISTEN)
mariadb     1060               mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1149 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1215 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1233 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1238 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1349 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
mariadb     1060 1352 mariadb     mysql   20u    IPv6        23706      0t0      TC
P *:mysql (LISTEN)
```

Монтирование NFS на клиенте

```
root@server:/etc
[root@server etc]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
Output information may be incomplete.
systemd      1          root    63u      IPv4        45663      0t0      UD
P *:sunrpc
systemd      1          root    65u      IPv6        45677      0t0      UD
P *:sunrpc
avahi-daemon 637        avahi    12u      IPv4        19843      0t0      UD
P *:mdns
avahi-daemon 637        avahi    13u      IPv6        19844      0t0      UD
P *:mdns
avahi-daemon 637        avahi    14u      IPv4        19845      0t0      UD
P *:60877
avahi-daemon 637        avahi    15u      IPv6        19846      0t0      UD
P *:52703
chronyd      658        chrony   5u       IPv4        19840      0t0      UD
P localhost:323
chronyd      658        chrony   6u      IPv6        19841      0t0      UD
P localhost:323
chronyd      658        chrony   7u      IPv4        19842      0t0      UD
P *:ntp
NetworkManager 928        root    30u      IPv4        22111      0t0      UD
P server.user.net:bootpc->_gateway:bootps
NetworkManager 928 931 gmain    root    30u      IPv4        22111      0t0      UD
P server.user.net:bootpc->_gateway:bootps
NetworkManager 928 932 gdbus    root    30u      IPv4        22111      0t0      UD
P server.user.net:bootpc->_gateway:bootps
rpcbind      7310       rpc      5u       IPv4        45663      0t0      UD
P *:sunrpc
rpcbind      7310       rpc      7u      IPv6        45677      0t0      UD
P *:sunrpc
rpc.statd    7313       rpcuser  7u      IPv4        54670      0t0      UD
P *:50942
rpc.statd    7313       rpcuser  9u      IPv4        54664      0t0      UD
P localhost:708
```

Автомонтирование через fstab на клиенте

```
[root@server etc]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit
ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin b
itcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine
checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp d
hcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync elasticse
arch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps fre
eipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-
availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isn
s jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver ku
be-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure
kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly k
ubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-
udp managesieve matrix mdns memcache minidlna mongod mosh mountd mqtt mqtt-tls ms-wbt mssql murmu
r mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imag
eio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresq
l privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetma
ster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba sa
mba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptra
p spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syn
cthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks tr
ansmission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discove
ry ws-discovery-client ws-discovery-tcp ws-discovery-udp wsmann wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
bash: firewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]#
```

Проверка remote-fs.target

```
[root@client ~]# mount server.user.net:/srv/nfs /mnt/nfs
mount: /mnt/nfs: bad option; for several filesystems (e.g. nfs, cifs) you might need a /sbin/mount
.<type> helper program.
[root@client ~]# mount
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,seclabel)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=4096k,nr_inodes=448710,mode=755,inode64)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,seclabel,gid=5,mode=620,ptmxmode=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,size=729744k,nr_inodes=819200,mode=755,inode64)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,seclabel,nsdelegate,memory
_recursiveprot)
pstore on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime,seclabel)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
/dev/sda1 on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)
selinuxfs on /sys/fs/selinux type selinuxfs (rw,nosuid,noexec,relatime)
systemd-1 on /proc/sys/fs/binfmt_misc type autofs (rw,relatime,fd=29,pgrp=1,timeout=0,minproto=5,m
axproto=5,direct,pipe_ino=19574)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime,seclabel)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,relatime,seclabel,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime,seclabel)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime,seclabel)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-sysctl.service type ramfs (ro,nosuid,nodev,noexec,relatime,seclab
el,mode=700)
```

Rocky

Экспорт каталога веб-сервера

```
p_id=1000)
[root@client ~]# suytemctl status remote-fs.target
bash: suytemctl: command not found...
[root@client ~]# systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
   Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset: enabled)
   Active: active since Thu 2026-03-12 10:45:13 UTC; 52min ago
     Until: Thu 2026-03-12 10:45:13 UTC; 52min ago
    Docs: man:systemd.special(7)

Mar 12 10:45:13 client.user.net systemd[1]: Reached target Remote File Systems.
[root@client ~]#
```

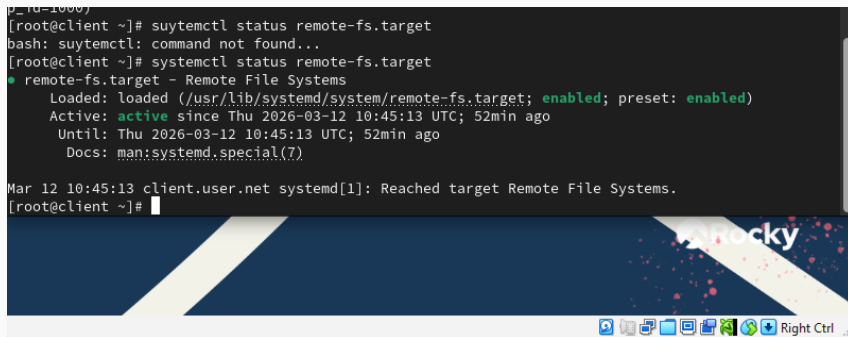


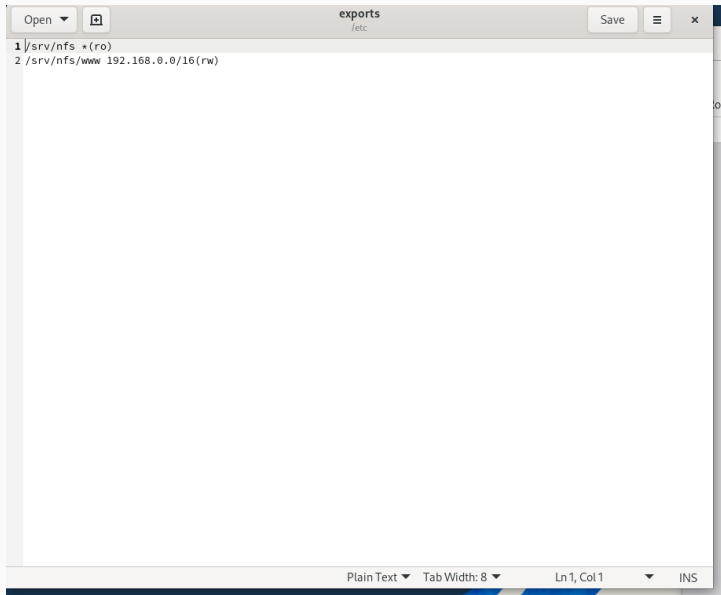
Рис. 8: Добавление экспорта `/srv/nfs/www` в `/etc/exports`

Bind-монтирование веб-каталога через fstab

```
l privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetma
ster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba sa
mba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptra
p spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syn
cthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks tr
ansmission-client upnp-client vdsu vnc-server warpinator wbem-http wbem-https wireguard ws-discove
ry ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
bash: firewall-cmd: command not found...
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[root@server etc]# firewall-cmd --add-service=mountd --add-service=rpc-bind --permanent
success
[root@server etc]# firewall-cmd --reload
success
[root@server etc]# mkdir -p /srv/nfs/www
[root@server etc]# mount -o bind /var/www/ /srv/nfs/www/
[root@server etc]#
```

Рис. 9: Запись bind-монтирования /var/www в /etc/fstab

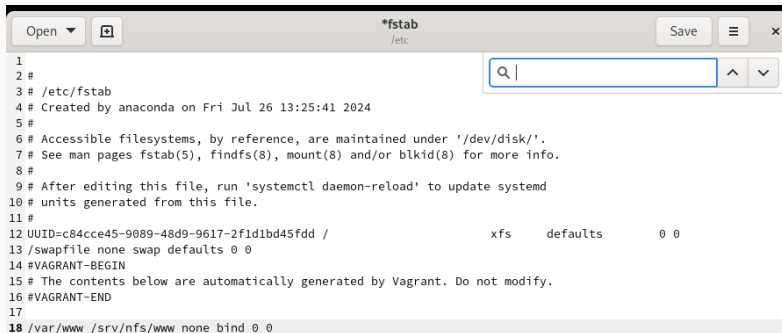
Проверка структуры дерева NFS на сервере



The image shows a text editor window titled "exports /etc". The editor contains two lines of text: "1 /srv/nfs *(ro)" and "2 /srv/nfs/www 192.168.0.0/16(rw)". The interface includes a top bar with "Open", "Save", and a menu icon, and a bottom status bar showing "Plain Text", "Tab Width: 8", "Ln 1, Col 1", and "INS".

```
1 /srv/nfs *(ro)
2 /srv/nfs/www 192.168.0.0/16(rw)
```

Проверка структуры дерева NFS на клиенте



The screenshot shows a text editor window titled `*fstab` with the path `/etc` below it. The window has a menu bar with `Open`, `Save`, and a hamburger menu icon. A search bar is located on the right side of the editor. The content of the file is as follows:

```
1
2 #
3 # /etc/fstab
4 # Created by anaconda on Fri Jul 26 13:25:41 2024
5 #
6 # Accessible filesystems, by reference, are maintained under '/dev/disk/'.
7 # See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
8 #
9 # After editing this file, run 'systemctl daemon-reload' to update systemd
10 # units generated from this file.
11 #
12 UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /          xfs      defaults    0 0
13 /swapfile none swap defaults 0 0
14 #VAGRANT-BEGIN
15 # The contents below are automatically generated by Vagrant. Do not modify.
16 #VAGRANT-END
17
18 /var/www /srv/nfs/www none bind 0 0
```

Подготовка пользовательского каталога

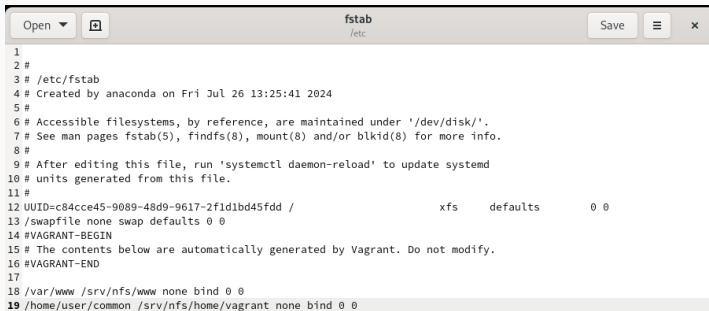
```
(gedit:7666): dconf-WARNING **: 12:08:58.809: failed to commit changes to dconf: Failed to execute
child process "dbus-launch" (No such file or directory)
[root@server etc]# exportfs -r
[root@server etc]# mkdir -p -m 700 ~/common
[root@server etc]# cd ~/common
[root@server common]# touch user@server.txt
[root@server common]# mkdir -p /srv/nfs/home/user
[root@server common]# mkdir -p /srv/nfs/home/vagrant
[root@server common]# mount -o bind /home/user/common /srv/nfs/home/vagrant
mount: /srv/nfs/home/vagrant: special device /home/user/common does not exist.
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[root@server common]#
```

Рис. 12: Создание каталога common и файла пользователя на сервере

Экспорт пользовательского каталога



Рис. 13: Добавление экспорта `/srv/nfs/home/akabrikosov` в `/etc/exports`



The screenshot shows a text editor window titled 'fstab /etc'. The window has a menu bar with 'Open', a file icon, and 'Save'. The content of the file is as follows:

```
1
2 #
3 # /etc/fstab
4 # Created by anaconda on Fri Jul 26 13:25:41 2024
5 #
6 # Accessible filesystems, by reference, are maintained under '/dev/disk/'.
7 # See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
8 #
9 # After editing this file, run 'systemctl daemon-reload' to update systemd
10 # units generated from this file.
11 #
12 UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /          xfs     defaults      0 0
13 /swapfile none swap defaults 0 0
14 #VAGRANT-BEGIN
15 # The contents below are automatically generated by Vagrant. Do not modify.
16 #VAGRANT-END
17
18 /var/www /srv/nfs/www none bind 0 0
19 /home/user/common /srv/nfs/home/vagrant none bind 0 0
```

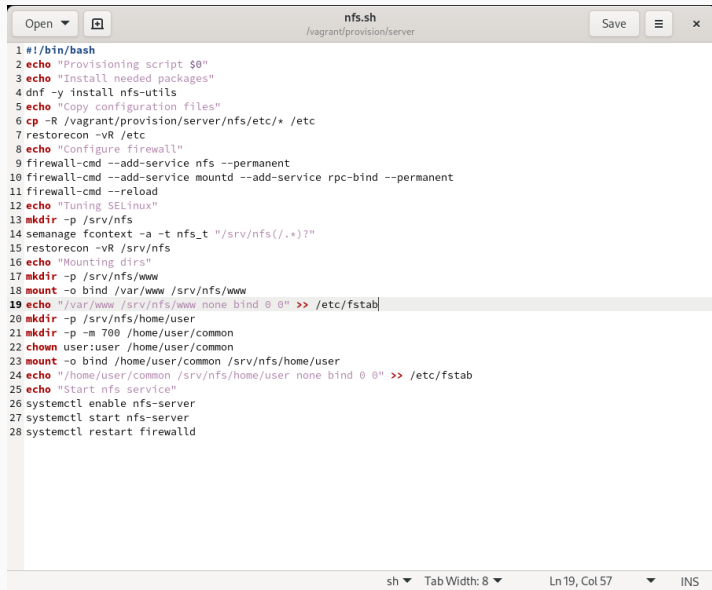
Проверка монтирования на сервере

```
[root@client ~]# suytemctl status remote-fs.target
bash: suytemctl: command not found...
[root@client ~]# systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
   Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset: enabled)
   Active: active since Thu 2026-03-12 10:45:13 UTC; 52min ago
     Until: Thu 2026-03-12 10:45:13 UTC; 52min ago
    Docs: man:systemd.special(7)

Mar 12 10:45:13 client.user.net systemd[1]: Reached target Remote File Systems.
[root@client ~]# cd /mnt/nfs/home/user
-bash: cd: /mnt/nfs/home/user: No such file or directory
[root@client ~]# cd /mnt/nfs/home
-bash: cd: /mnt/nfs/home: No such file or directory
[root@client ~]# mkdir /mnt/nfs/home/user
mkdir: cannot create directory '/mnt/nfs/home/user': No such file or directory
[root@client ~]# mkdir -p /mnt/nfs/home/user
[root@client ~]# cd /mnt/nfs/home/user
[root@client user]# touch user@client.txt
[root@client user]#
```

Рис. 15: Проверка активных монтирований /srv/nfs/www и /srv/nfs/home/akabrikosov

Проверка прав доступа на клиенте

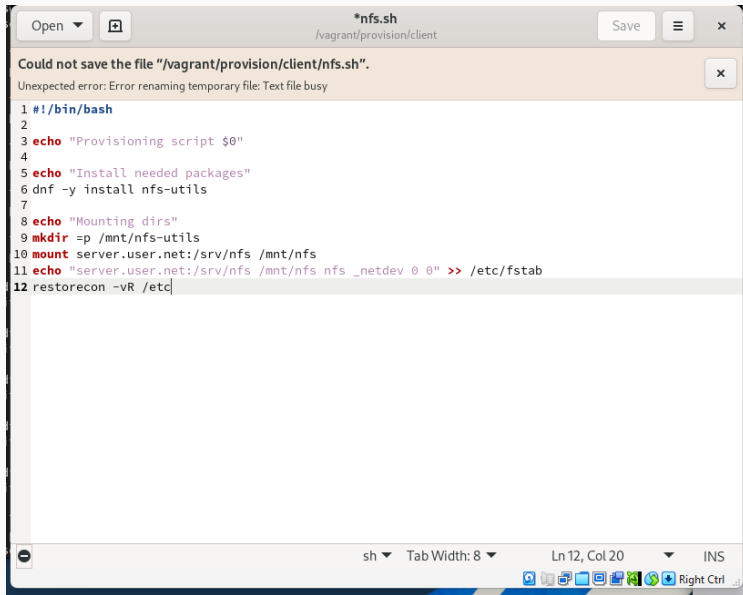


```
nfs.sh
/vagrant/provision/server

1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install nfs-utils
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/nfs/etc/* /etc
7 restorecon -vR /etc
8 echo "Configure firewall"
9 firewall-cmd --add-service nfs --permanent
10 firewall-cmd --add-service mountd --add-service rpc-bind --permanent
11 firewall-cmd --reload
12 echo "Tuning SELinux"
13 mkdir -p /srv/nfs
14 semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
15 restorecon -vR /srv/nfs
16 echo "Mounting dirs"
17 mkdir -p /srv/nfs/www
18 mount -o bind /var/www /srv/nfs/www
19 echo "/var/www /srv/nfs/www none bind 0 0" >> /etc/fstab
20 mkdir -p /srv/nfs/home/user
21 mkdir -p -m 700 /home/user/common
22 chown user:user /home/user/common
23 mount -o bind /home/user/common /srv/nfs/home/user
24 echo "/home/user/common /srv/nfs/home/user none bind 0 0" >> /etc/fstab
25 echo "Start nfs service"
26 systemctl enable nfs-server
27 systemctl start nfs-server
28 systemctl restart firewalld
```

sh Tab Width: 8 Ln 19, Col 57 INS

Проверка синхронизации данных на сервере



The screenshot shows a Vagrant terminal window with a title bar containing "Open", a file icon, "*nfs.sh", and the path "/vagrant/provision/client". On the right side of the title bar are "Save", a menu icon, and a close button. An orange error banner at the top reads: "Could not save the file "/vagrant/provision/client/nfs.sh". Unexpected error: Error renaming temporary file: Text file busy". Below the banner is a script with 12 lines of code. The terminal status bar at the bottom shows "sh", "Tab Width: 8", "Ln 12, Col 20", and "INS". The system taskbar at the very bottom includes icons for applications and system status, with "Right Ctrl" visible.

```
1 #!/bin/bash
2
3 echo "Provisioning script $0"
4
5 echo "Install needed packages"
6 dnf -y install nfs-utils
7
8 echo "Mounting dirs"
9 mkdir -p /mnt/nfs-utils
10 mount server.user.net:/srv/nfs /mnt/nfs
11 echo "server.user.net:/srv/nfs /mnt/nfs nfs _netdev 0 0" >> /etc/fstab
12 restorecon -vR /etc
```

Выводы по проделанной работе

В ходе лабораторной работы настроены сервер и клиент NFSv4. Реализовано экспортирование каталогов с различными уровнями доступа: общий каталог, каталог веб-сервера и пользовательский каталог.

Проверена корректность bind-монтирования, ограничений прав доступа и автоматического подключения сетевых ресурсов при загрузке системы.

Настройка окружения автоматизирована с помощью provision-скриптов Vagrant, что обеспечивает воспроизводимость развертывания.